Fall 2026 ECE 445 Team Contract

Instructions: The content of this document should be specific to your goals and needs. Ideas for the content of each section are provided as suggestions.

Project No. and Name	
Member Name, netID	Ben Wang, bw34
Member Name, netID	Daniel Ly, dhly2
Member Name, netID	Hyun Been Park (Robin), hbpark2

ECE 445 is a project-based course. The course includes both team and individual grades. Project teammates generally all get the same grade for team assignments based on the expectation that all team members do their fair share of the work involved. The purpose of this contract is to lay out the tasks needed for the successful completion of the project and distribute them in a fair and efficient way to the team members. It will also discuss how the teammates will work together during the project and address any issues that come up. A contract that promotes good teamwork that leads to a successful project should:

- Acknowledge that each team member has commitments and responsibilities outside of ECE 445
- Encourage open communication about challenges that team members are facing, both in and out of ECE 445
- Give team members the benefit of the doubt and the opportunity to explain themselves when something goes wrong and resist jumping to judgement

Project Description: *Short description of project*

Our Senior Design project is titled TROLL E: Tracked Robotic Outdoor Load-Lugging Escort. A play on the character WALL-E, this project is intended to build a trolley-like vehicle, able to carry up to 20lbs of load, to assist in everyday outings where someone may need to carry something with them. To accomplish this, we will need to design the frame of the cart, along with integrating all necessary sensors and their method of communication to the central microcontroller. We will focus on protective measures to ensure that TROLL E can navigate itself through different obstacles and terrain, as well as general functionality like following the user.

Project Goals: *If the team is successful in its purpose, what hardware and software achievements will attest to this?*

Some hardware achievements that will show our success include proper integration of each sensor and their ability to communicate with the central microcontroller. This means that the LiDAR and ultrasonic

sensors are collecting data and informing the MCU so that it can make accurate decisions on pathing and obstacle detection. The weight sensor and Ultra-Wideband will also show functionality when they work as intended. Additionally, functionality in the motors and wheels is another indicator of proper implementation and can show our success in the hardware aspects of this project.

For the software achievements that can help show our success, we are looking at the decision-making of TROLL E after receiving information from each of its sensors. This will vary from the LiDAR signal influencing the general path-finding of TROLL E, to the ultrasonic sensors providing a more general failsafe for pathing against smaller objects, along with both the weight sensor and Ultra-Wideband signal giving emergency stop signals under their edge conditions. Furthermore, we can test software implementation by observing how TROLL E utilizes the Mecanum wheels for omnidirectional movement. Because of the wheel design, TROLL E will be able to make more precise movements, which depend on how we program the motors to spin depending on the overall path that was found after interpreting all data collected.

Expectations (ground rules) for each member: *Try to list six or more minimum expectations. Consider aspects such as preparation, participation, feedback, responsiveness, etc. Try to explicitly list anything that could potentially turn into a problem. Find ways to encourage everyone to communicate (this may also fall under "tasks").*

- Stay up to date with all discussions that occur, whether they be through our Discord chat or in person. This will mean that any discussion must be available to any member that did not participate in the discussion. The other members will need to assist to the best of their ability to bring their partner up to speed.
- During discussions, be courteous and allow others to speak their thoughts to get the full picture of what everyone is thinking. This is meant to let everyone contribute to the discussions, and so that no one will be left out.
- With PCB design, whoever is the main designer or last made changes to the design should let the rest of the team know via Discord or another method of communication. This will help keep everyone on the same page.
- Before any critical deadlines, the whole team will meet up in person in a location determined at the time, so that we can all discuss anything that needs to be changed last second or if the portion of our project is good.
- For the overall project design, key decisions will be made as a team, which includes what devices we plan to use, ordering anything online, if we plan to use certain equipment not previously mentioned in our proposal, or anything along these lines. No singular member will act on their own in this respect, and will need to talk to at least one other member, ideally both.

Roles: *Do you see this team performing well because everyone works together and contributes equally? Are there certain aspects of the project that some teammates excel at? Can tasks be spread among individuals to optimize progress toward the final product?*

Ben: Will focus on sensors and microcontroller

Daniel: Will focus on motors and battery systems

Robin: Will focus on sensors, but subject to change should Daniel request help

The focuses are set this way based on how much prior knowledge we all have, but this doesn't mean we will be siloed for the duration of our project. We intend to have each member understand the total functionality of each subsystem, with the member focusing on it being the 'expert' in the team.

Project Meeting Time(s): *The team will meet at the scheduled team meeting with TA each week. Can you also preset an ideal time for team meetings in the lab (your team may need to sign up for lab bench access)? Is your team interested in meeting to work on other aspects of the course together such as project research?*

Weekly meetings with the TA are currently scheduled for 4:15pm on Tuesday, but can be subject to change should anyone in the team or TA provide notice in advance, with a cutoff of 24 hours.

For meeting in the lab, we can use the time after our weekly meetings with our TA, but will prioritize Thursday as a lab work day. This is heavily subject to change as we understand schedules can change, so should anything conflict with this day, we will discuss it together and find other times to meet in the lab. This doesn't need to be a full team meeting either, so if any pair wishes to go to the lab while the third has another commitment, the pair can meet and will document their work and inform the third about what progress was made.

Agenda: *Who will set the agenda? Beyond the weekly meetings with the TA, what will the team do to ensure that it stays on track during the semester? When a decision needs to be made, will it be approved by consensus or majority vote? Will a team member be appointed to keep records?*

Our agendas are very flexible and will be determined by each member weighing in on what our priorities are during each team meeting or from prior discussions. These are also able to branch out so that each team member is progressing their own research or PCB design, so we think it makes more sense to give each other more freedom to do what they think is best, while still communicating what they plan on doing. Should anything critical come up, we think it's best to come to a majority vote on what we think we should do.

Process and penalties for dealing with team issues: What happens when ground rules are broken? Who intervenes? What happens if the situation escalates? Always remember not to jump to judgement. Give group members the benefit of the doubt and the opportunity to explain themselves when something first goes wrong. TAs and instructors are available to help resolve issues.

With our methods of communication established, it should be very easy for any one member to reach out to another and talk about anything that may have come up at the time. In the unlikely event that a situation arises and/or escalates, we plan to come together as a team for an impromptu meeting to discuss the matter. So far, we have been giving good notice on our responsibilities outside this class on top of what work we want to complete, so we've already established a good foundation of how to talk to each other.

End-of-term agreement on using final peer assessment for grade adjustment: Do you believe that this contract should hold your team accountable to its contents or that it may hold little value? There will be two formal peer assessments this semester. The first is used only to provide honest, constructive feedback to each team member. The second peer assessment affects a teammate's grade. Without accountability, many promises go by the wayside.

This contract should be a foundational item for how we evaluate our teammates, but shouldn't be the driving force behind how we assess each other. Using this as a general guideline for how each team member should contribute to the project as a whole, we can make informed decisions later based on what we've established here. However, situations are prone to change, so we should consider that anyone on the team could need to change their contribution amount for unforeseen circumstances, which will be discussed as a team. So long as all complications are professionally and respectfully discussed, final peer assessments should reflect the positive teamwork and environment we've seen so far in the course.

Signatures: Iterate on this document until everyone is comfortable with its contents and signs (it is okay to type your printed name as your digital signature).

I affirm that I participated in generating this team charter and that I will abide by its contents to the best of my ability. Furthermore, I understand that failure to meet the expectations expressed here can lead to the stated consequences.

netID: _____bw34_____ (digital) Signature: ____Benjamin Wang____ Date: 9/23/26

netID: _____dhly2_____ (digital) Signature: _____Daniel Ly_____ Date: 9/23/26

netID: _____hbpark2___ (digital) Signature: _Hyun Been Park_____ Date: 9/24/26